Lattice QCD Analysis Methods

HMC GPU Simulation

March 18, 2026

1. Autocorrelation Function

For a sequence of measurements $\{O_0, O_1, O_2, \ldots, O_{N-1}\}$, the autocorrelation function is defined as:

$$\Gamma(t) = \frac{1}{N-t} \sum_{i=0}^{N-t-1} (O_i - \langle O \rangle)(O_{i+t} - \langle O \rangle) \tag{1}$$

where the average is

$$\langle O \rangle = \frac{1}{N} \sum_{i=0}^{N-1} O_i \tag{2}$$

2. Normalized Autocorrelation Function

The normalized autocorrelation function is:

$$\rho(t) = \frac{\Gamma(t)}{\Gamma(0)} \tag{3}$$

where $\Gamma(0)$ is the variance of the data:

$$\Gamma(0) = \mathrm{Var}(O) = \frac{1}{N} \sum_{i=0}^{N-1} (O_i - \langle O \rangle)^2 \tag{4}$$

3. Integrated Autocorrelation Time

The integrated autocorrelation time characterizes the number of effectively independent measurements:

$$\tau_{\mathrm{int}} = \frac{1}{2} + \sum_{t=1}^{W} \rho(t) \tag{5}$$

where W is the cutoff window. In our analysis, we use $W = N/2$.

4. Statistical Error

The error on the mean, accounting for autocorrelation, is:

$$\sigma_{\langle O \rangle} = \sqrt{\frac{2\tau_{\text{int}}}{N}}\sigma_O \tag{6}$$

where $\sigma_O = \sqrt{\Gamma(0)}$ is the standard deviation.

5. Plaquette Observable

The plaquette is the fundamental Wilson loop around a single plaquette:

$$P_{\mu\nu}(x) = \text{Tr}\left[U_\mu(x)U_\nu(x+\hat{\mu})U_\mu^\dagger(x+\hat{\nu})U_\nu^\dagger(x)\right] \tag{7}$$

The average plaquette:

$$\langle P \rangle = \frac{1}{N_t N_s^3 \cdot 6 \cdot 3}\sum_x \sum_{\mu<\nu} \text{Re}\left[P_{\mu\nu}(x)\right] \tag{8}$$

6. Topological Charge

The topological charge is computed using the clover field strength tensor:

$$Q = \frac{1}{32\pi^2}\sum_x \sum_{\mu\nu\rho\sigma} \varepsilon_{\mu\nu\rho\sigma}\text{Tr}\left[F_{\mu\nu}(x)F_{\rho\sigma}(x)\right] \tag{9}$$

The field strength tensor is constructed from 4 oriented plaquettes at each site:

$$C_{\mu\nu}(x) = \frac{1}{4}\left[P_{\mu\nu}(x) + P_{\nu,-\mu}(x) + P_{-\mu,-\nu}(x) + P_{-\nu,\mu}(x)\right] \tag{10}$$

$$F_{\mu\nu}(x) = \frac{1}{2}(C_{\mu\nu} - C_{\mu\nu}^\dagger)_{traceless} \tag{11}$$

7. Gauge Field Smearing

In lattice QCD, gauge field smearing is a technique used to reduce short-distance noise in gauge configurations while preserving long-range topological structures. Smearing iteratively replaces each link variable with a weighted sum of itself and neighboring links, typically using a "staple" - a product of three links that forms the shortest path connecting the endpoints of the original link in a different direction. Below we describe five commonly used smearing algorithms: APE, STOUT, Over-Improved STOUT, HYP, and Wilson/Symanzik Flow.

7.1 APE Smearing

APE smearing (named after the APE collaboration) is one of the earliest and most straightforward smearing methods. It projects the link onto the SU(3) group using a linear combination of the original link and a weighted staple sum.

For each direction μ at site x, the updated link is computed as:

$$U'_\mu(x) = \mathcal{P}\left[(1-\alpha)U_\mu(x) + \frac{\alpha}{2(d-1)}\Sigma_\mu(x)\right] \tag{12}$$

where $d = 4$ is the spacetime dimension, α is the smearing parameter (typically $0.1 - 1.0$), and $\Sigma_\mu(x)$ is the staple sum:

$$\Sigma_\mu(x) = \sum_{\nu\neq\mu}\left[U_\nu(x)U_\mu(x+\hat{\nu})U^\dagger_\nu(x+\hat{\mu}) + U^\dagger_\nu(x-\hat{\nu})U_\mu(x-\hat{\nu})U_\nu(x-\hat{\nu}+\hat{\mu})\right] \tag{13}$$

The symbol $\mathcal{P}$ denotes projection back to the SU(3) group, typically performed using polar decomposition or Gram-Schmidt orthonormalization.

APE smearing is simple and computationally inexpensive but tends to suppress short-range fluctuations while allowing topological charge to evolve slowly.

7.2 STOUT Smearing

STOUT smearing (derived from "STOchastic UpdATe" or "STretching") uses an exponential mapping to update links, which better preserves the unitarity of the gauge field. Rather than a linear projection, STOUT applies an exponential factor to the link.

The update formula is:

$$U'_\mu(x) = \exp\left(i\rho\, Q_\mu(x)\right) U_\mu(x) \tag{14}$$

where ρ is the smearing parameter (typically $0.1 - 0.2$) and $Q_\mu(x)$ is a Hermitian matrix constructed from the staple:

$$\Omega_\mu(x) = \rho\, \Sigma_\mu(x) U^\dagger_\mu(x) \tag{15}$$

$$Q_\mu(x) = \frac{i}{2}\left(\Omega_\mu(x) - \Omega^\dagger_\mu(x) - \frac{1}{3}\mathrm{Tr}\left[\Omega_\mu(x) - \Omega^\dagger_\mu(x)\right]\mathbf{1}\right) \tag{16}$$

The exponential $\exp(i\rho Q_\mu)$ is computed using Taylor expansion or Cayley-Hamilton theorem, and the result is automatically unitary. The traceless Hermitian matrix Q_μ ensures the exponential stays in the SU(3) group.

STOUT smearing provides a more unitarity-preserving update than APE and is widely used in modern lattice QCD simulations.

7.3 Over-Improved STOUT Smearing

Over-Improved STOUT (OvrImpSTOUT) extends STOUT smearing by including longer "rectangle" terms (paths of length 5) in addition to the standard staples. This reduces the anisotropy in the smearing procedure and improves the accessibility of topological charge.

The update uses:

$$U'_\mu(x) = \exp\left(i\, Q_\mu^{\text{imp}}(x)\right) U_\mu(x) \tag{17}$$

where the improved Hermitian matrix is:

$$Q_\mu^{\text{imp}} = \frac{5}{3}\rho\, Q_\mu^{\text{stap}} - \frac{1}{12}\rho\, Q_\mu^{\text{rect}} \tag{18}$$

The staple and rectangle contributions are:

$$Q_\mu^{\text{stap}} = \left(\Sigma_\mu(x) U_\mu^\dagger(x)\right)_{\text{Hermitian}} \tag{19}$$

$$Q_\mu^{\text{rect}} = \left(R_\mu(x) U_\mu^\dagger(x)\right)_{\text{Hermitian}} \tag{20}$$

where $R_\mu(x)$ denotes the sum of rectangular loops of length 5 that connect the endpoints of the link $U_\mu(x)$.

The parameter ϵ (typically 0 to 0.25) controls the rectangle contribution. Setting $\epsilon = 0$ recovers standard STOUT, while $\epsilon = 1/8$ provides optimal rotational symmetry.

Over-Improved STOUT is particularly useful for studying topological charge because it allows faster tunneling between topological sectors compared to standard STOUT.

7.4 HYP Smearing

HYP smearing (named after Hasenfratz, Kuti, and others, or "HYPERcubic") uses a recursive three-level procedure based on hypercubic blocking. It is more sophisticated than APE or STOUT and provides excellent smoothing of short-distance fluctuations.

HYP smearing proceeds in three levels:

Level 1: For each link, compute the sum of all two-link paths connecting its endpoints (first HYP staple):

$$V_{\mu\nu}^{(1)}(x) = U_\nu(x) U_\mu(x+\hat{\nu}) U_\nu^\dagger(x+\hat{\mu}) + U_\nu^\dagger(x-\hat{\nu}) U_\mu(x-\hat{\nu}) U_\nu(x-\hat{\nu}+\hat{\mu}) \tag{21}$$

Level 2: Build three-link objects from the Level 1 links (second HYP staple):

$$V_{\mu\nu}^{(2)}(x) = \sum_{\rho\neq\mu,\nu} V_{\rho\nu}^{(1)}(x) V_{\mu\rho}^{(1)}(x+\hat{\nu}) (V_{\rho\nu}^{(1)}(x+\hat{\mu}))^\dagger \tag{22}$$

Level 3: The final link is updated as:

$$U'_\mu(x) = \mathcal{P}\left[U_\mu(x) + \alpha_1 \Sigma_\mu^{(1)}(x) + \alpha_2 \Sigma_\mu^{(2)}(x)\right] \tag{23}$$

where $\Sigma_\mu^{(1)}$ and $\Sigma_\mu^{(2)}$ are constructed using the Level 1 and Level 2 objects, respectively. Typical parameter values are $\alpha_1 = 0.75$, $\alpha_2 = 0.6$, and $\alpha_3 = 0.3$ (for 3D HYP).

HYP smearing is particularly effective at removing UV fluctuations while preserving the topological structure, making it popular for quark smearing and fat link constructions.

7.5 Wilson Flow and Symanzik Flow

Wilson Flow (and its Symanzik-improved variant) is not a traditional smearing method but rather a continuous evolution of the gauge field in a fictitious "flow time" dimension. It solves a differential equation that smooths the gauge field while maintaining exact gauge invariance.

The Wilson Flow is defined by the differential equation:

$$\frac{dU_\mu(x,t)}{dt} = -\langle G_\mu(x,t)\rangle U_\mu(x,t) \tag{24}$$

where t is the flow time and G_μ is the gauge force derived from the Wilson action. The initial condition is $U_\mu(x,0) = U_\mu(x)$ (the original gauge field).

For the standard Wilson flow (using 1×1 plaquettes), the force is:

$$G_\mu(x) = \sum_{\nu\neq\mu} [P_{\mu\nu}(x) - P_{\mu\nu}(x-\hat{\nu})] \tag{25}$$

where $P_{\mu\nu}(x)$ is the standard plaquette.

For Symanzik-improved flow (2×1 loop improvement), the force includes both single-plaquette and rectangular terms:

$$G_\mu^{\text{Sym}}(x) = c_1 \sum_\nu [P_{\mu\nu}(x) - P_{\mu\nu}(x-\hat{\nu})] + c_2 \sum_{\nu,\rho\neq\mu} [R_{\mu\nu\rho}(x) - R_{\mu\nu\rho}(x-\hat{\rho})] \tag{26}$$

with $c_1 = 5/3$ and $c_2 = 1/12$ for optimal Symanzik improvement.

The flow is discretized using a Runge-Kutta integrator with step size ϵ:

$$U_\mu(x,t+\epsilon) = \exp\left(\epsilon \langle G_\mu(x,t)\rangle\right) U_\mu(x,t) \tag{27}$$

Wilson Flow has several advantages:

- It provides a continuous definition of smoothing with a physical scale
- The field strength $F_{\mu\nu}$ can be measured at any flow time
- The scale w_0 (or other gradient flow scales) provides a physical reference scale
- Topological charge evolves smoothly with flow time

The topological charge can be measured directly from the flowed field strength:

$$Q(t) = \frac{1}{32\pi^2} \sum_x \sum_{\mu\nu\rho\sigma} \varepsilon_{\mu\nu\rho\sigma} \text{Tr}\left[F_{\mu\nu}(x,t) F_{\rho\sigma}(x,t)\right] \tag{28}$$

Wilson Flow is widely used for scale setting, topology studies, and as a preconditioning step in dynamical QCD simulations.

Table 1: Comparison of gauge smearing methods

Method	Update Type	Parameters	Main Use Case
APE	Linear projection	α	Basic smoothing, fat links
STOUT	Exponential	ρ	Topology, improved actions
OvrImpSTOUT	Exponential	ρ, ϵ	Topology, fast tunneling
HYP	Multi-level projection	$\alpha_1, \alpha_2, \alpha_3$	Quark smearing, cooling
Wilson Flow	Continuous flow	ϵ	Scale setting, topology

14. Gradient Flow and Scale Setting

The gradient flow provides a way to define a continuous scale on the lattice.

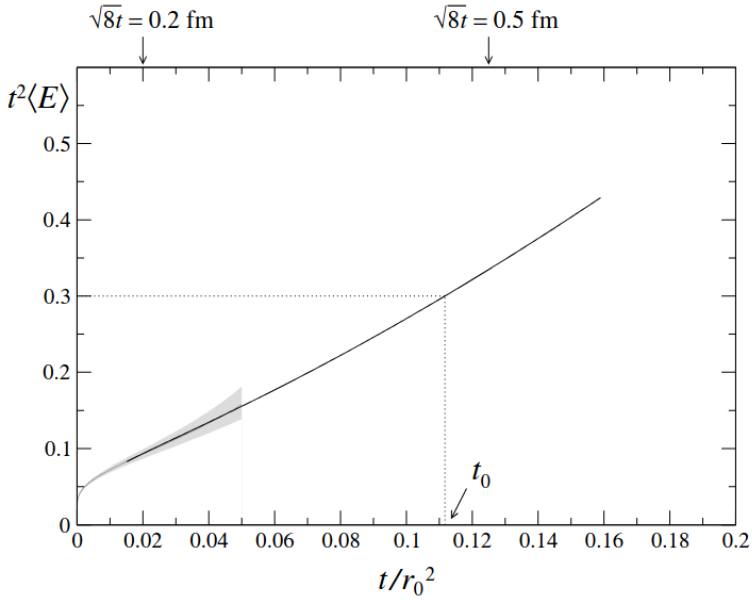


Figure 1: Gradient flow energy density from Luscher (2010).

Lattice Size Comparison at $\beta = 5.7$

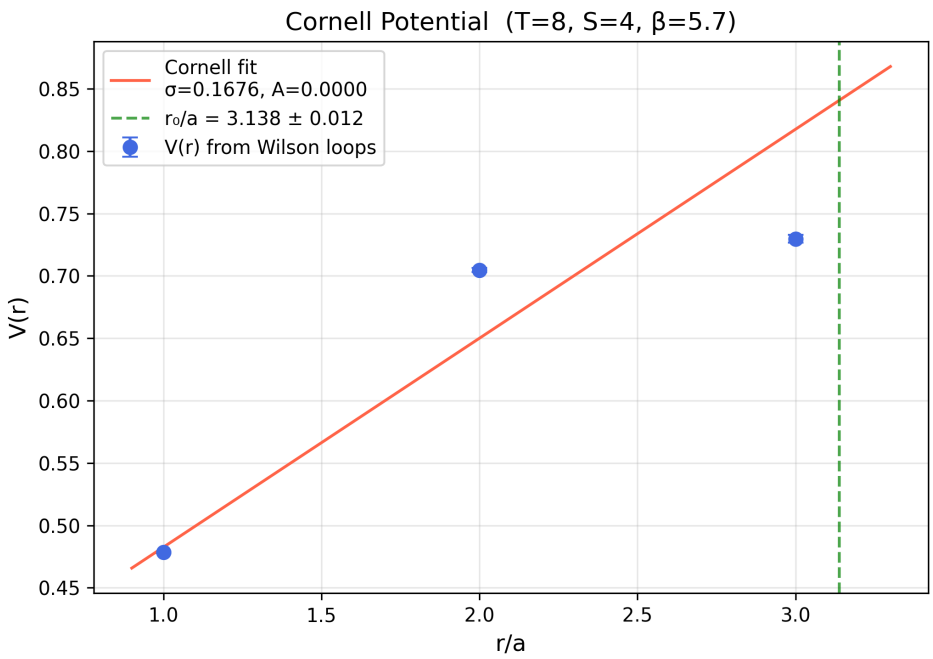


Figure 2: Cornell potential fit for 8×4^3 at $\beta = 5.7$

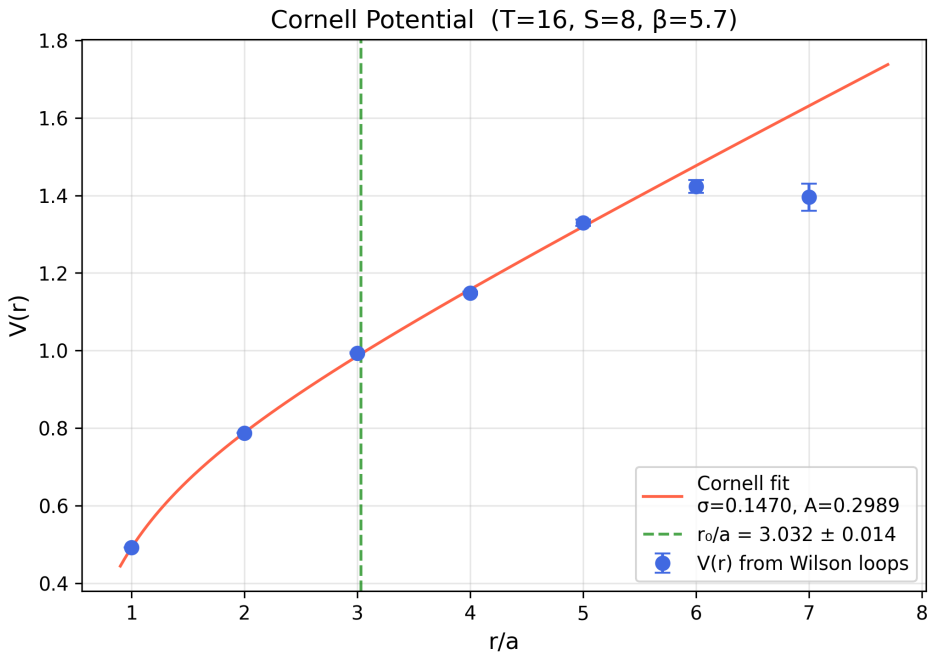


Figure 3: Cornell potential fit for 16×8^3 at $\beta = 5.7$

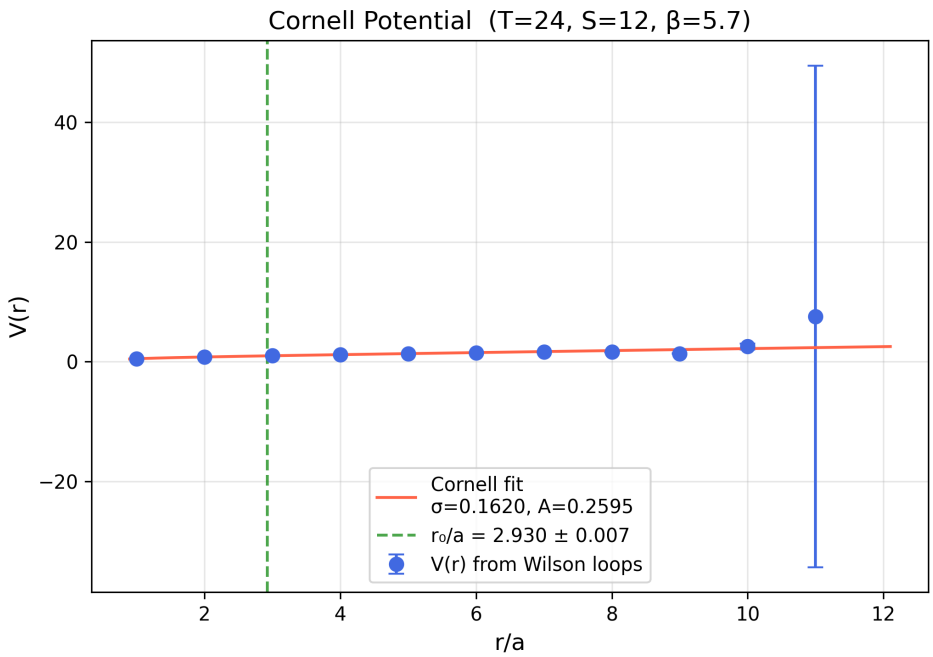


Figure 4: Cornell potential fit for 24×12^3 at $\beta = 5.7$

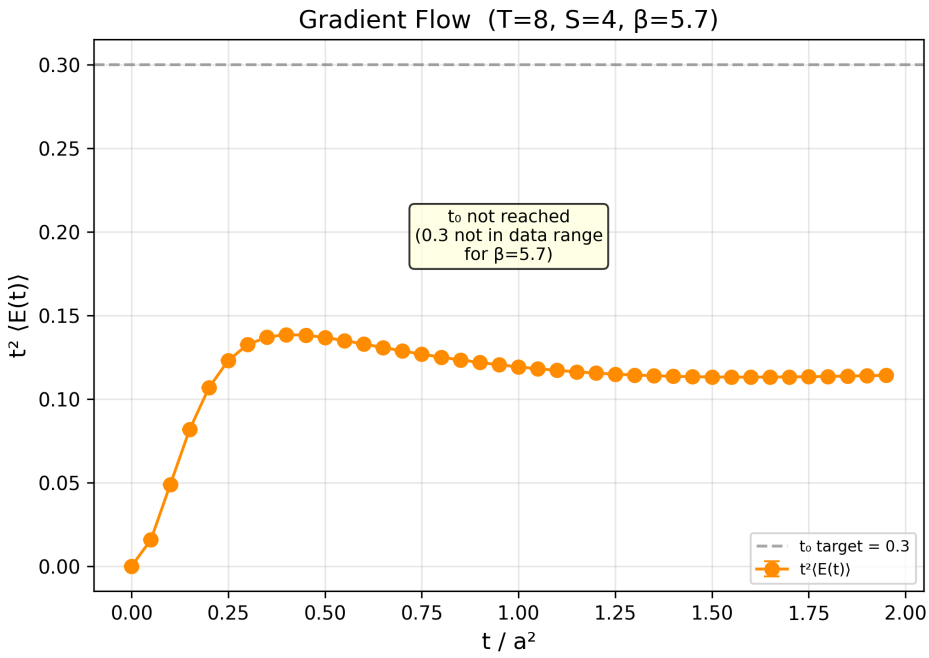


Figure 5: Gradient flow scale for 8×4^3 at $\beta = 5.7$

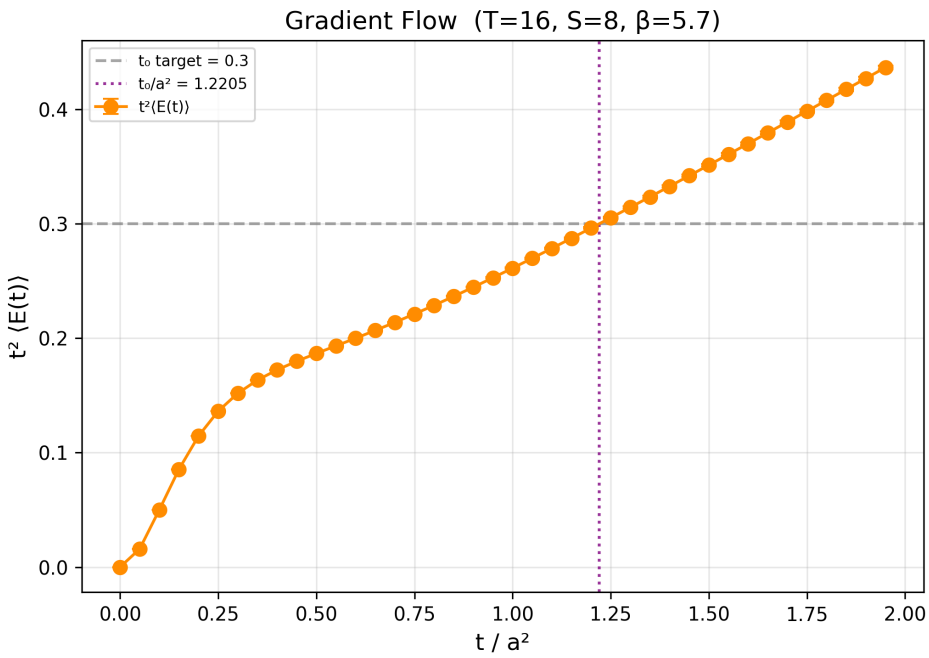


Figure 6: Gradient flow scale for 16×8^3 at $\beta = 5.7$

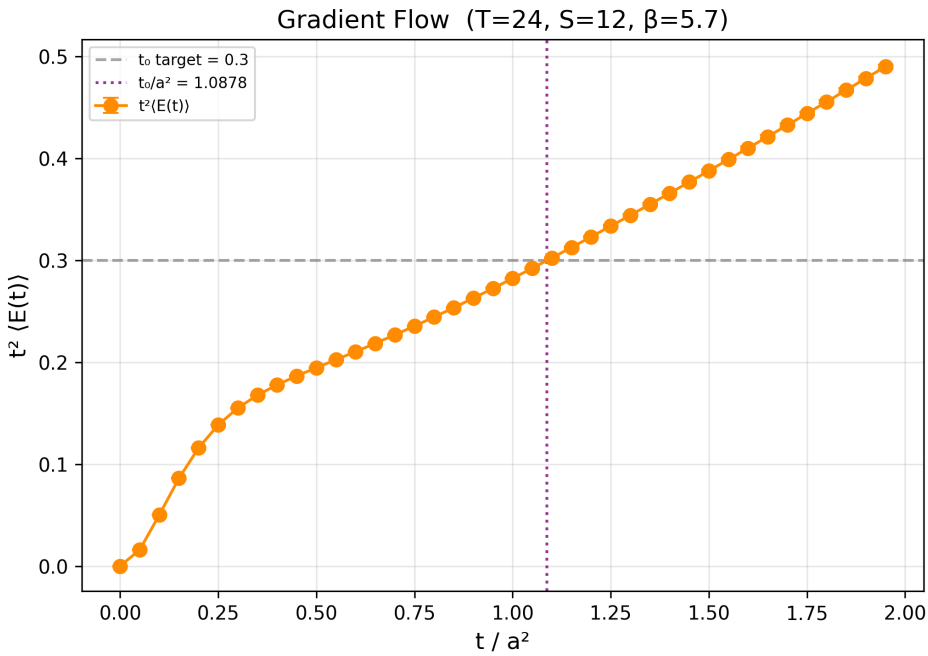


Figure 7: Gradient flow scale for 24×12^3 at $\beta = 5.7$

Beta Dependence at Fixed Volume (16×8^3)

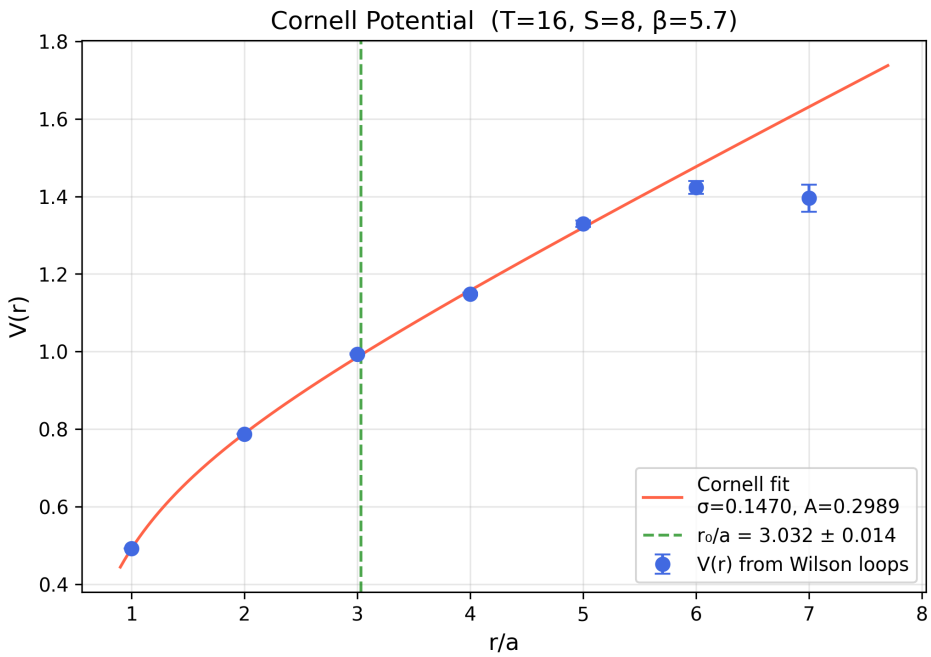


Figure 8: Cornell potential fit for $\beta = 5.7$

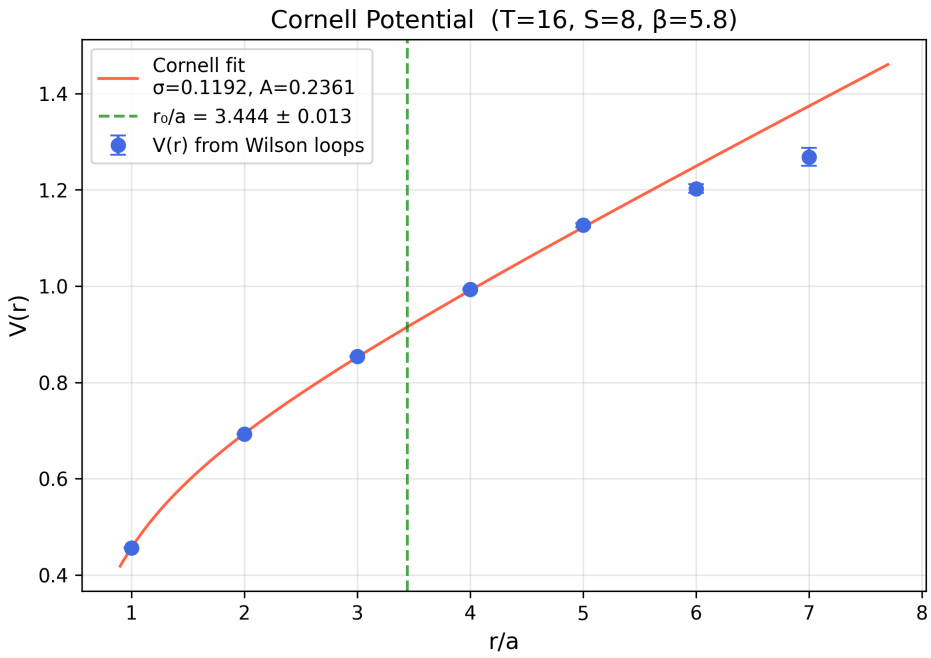


Figure 9: Cornell potential fit for $\beta = 5.8$

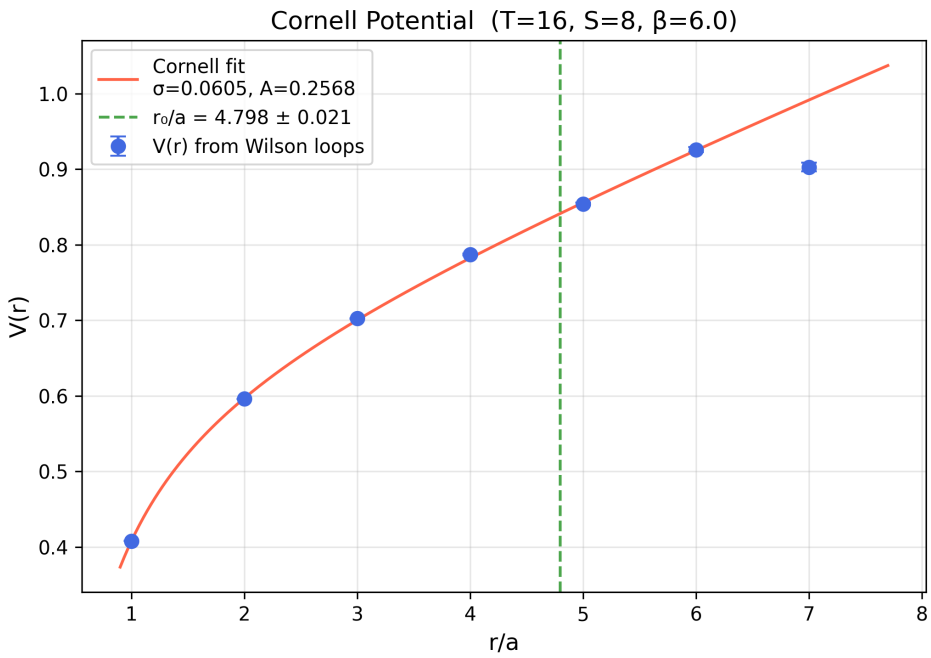


Figure 10: Cornell potential fit for $\beta = 6.0$

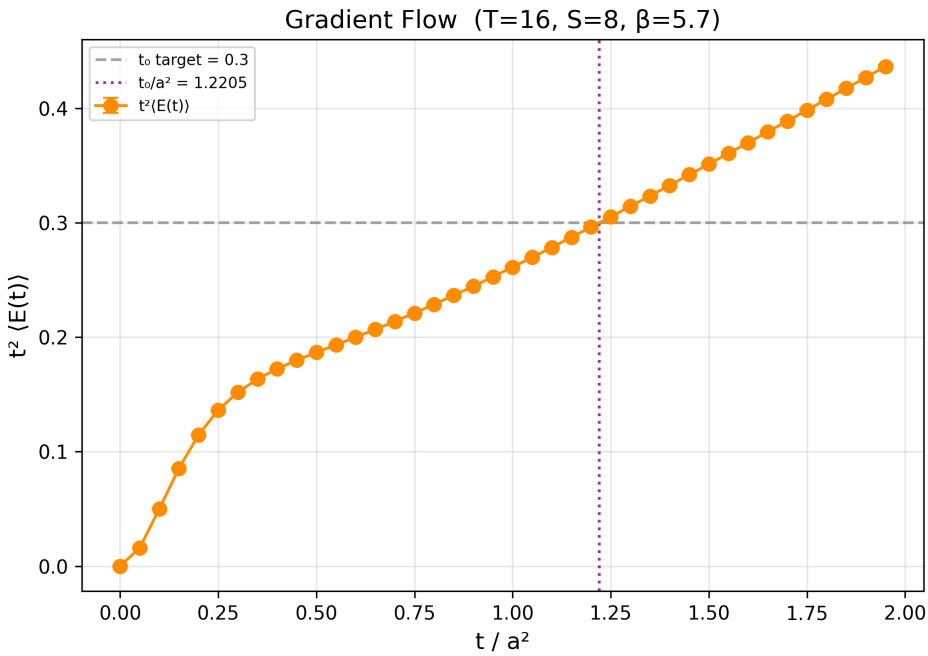


Figure 11: Gradient flow scale for $\beta = 5.7$

As expected, increasing β decreases the lattice spacing a.

Topological Charge Evolution

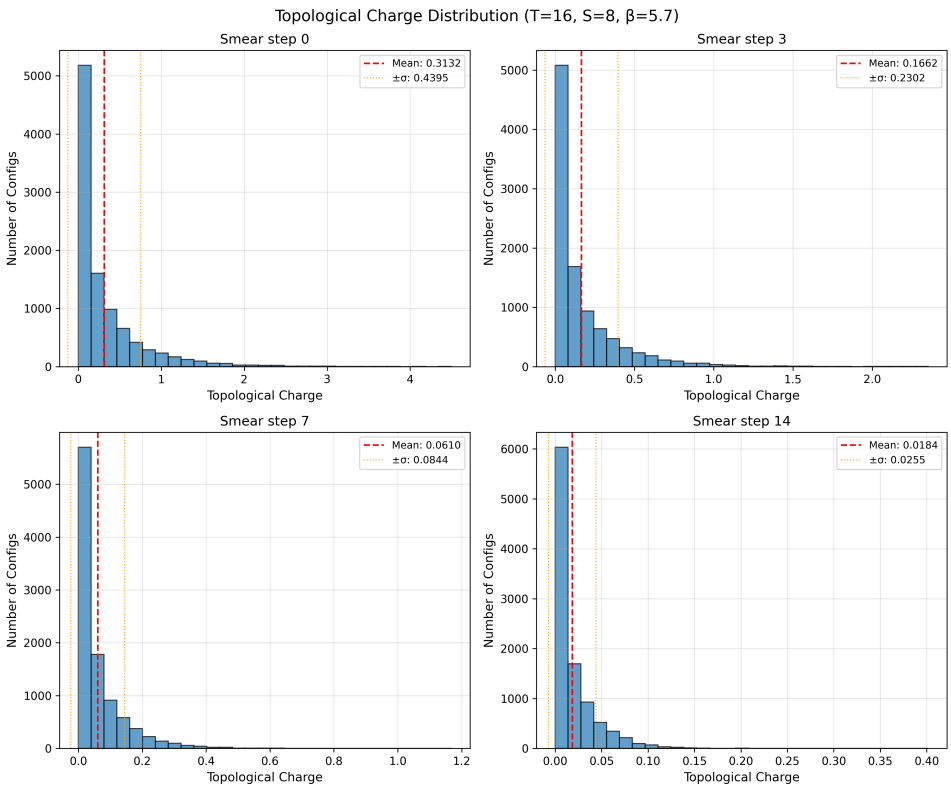


Figure 12: Topological charge Q^2 distribution for 16×8^3 at $\beta = 5.7$

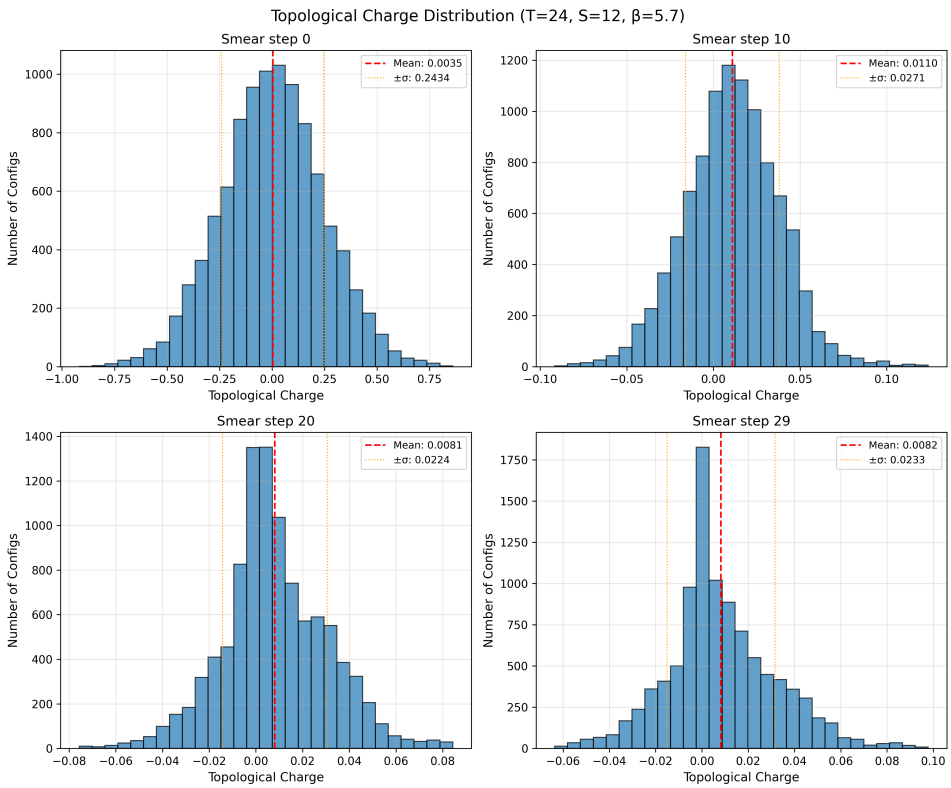


Figure 13: Topological charge Q^2 distribution for 24×12^3 at $\beta = 5.7$

Lattice Size Comparison at $\beta = 5.7$

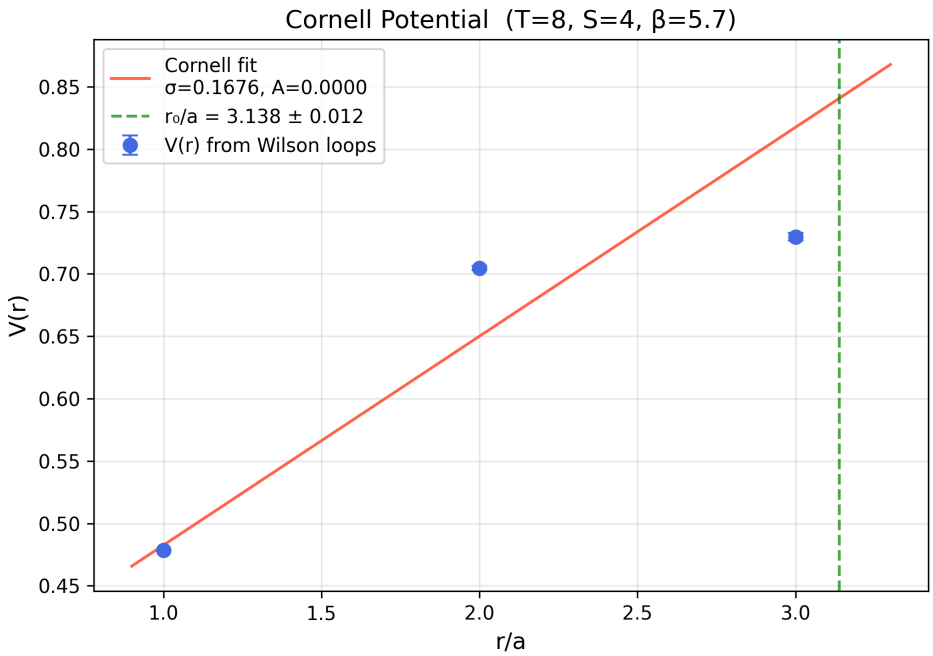


Figure 14: Cornell potential fit for 8×4^3 at $\beta = 5.7$

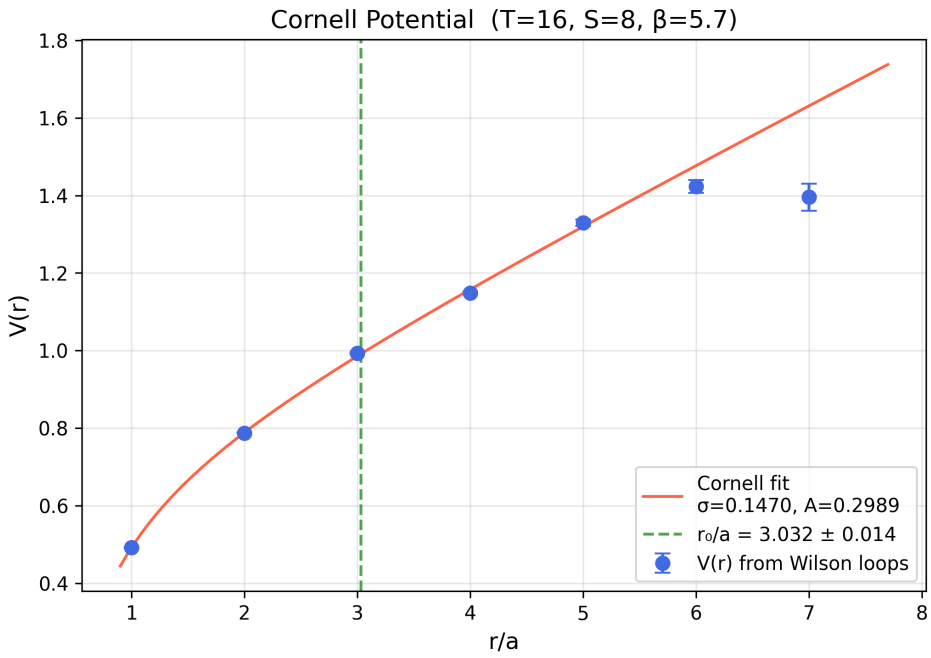


Figure 15: Cornell potential fit for 16×8^3 at $\beta = 5.7$

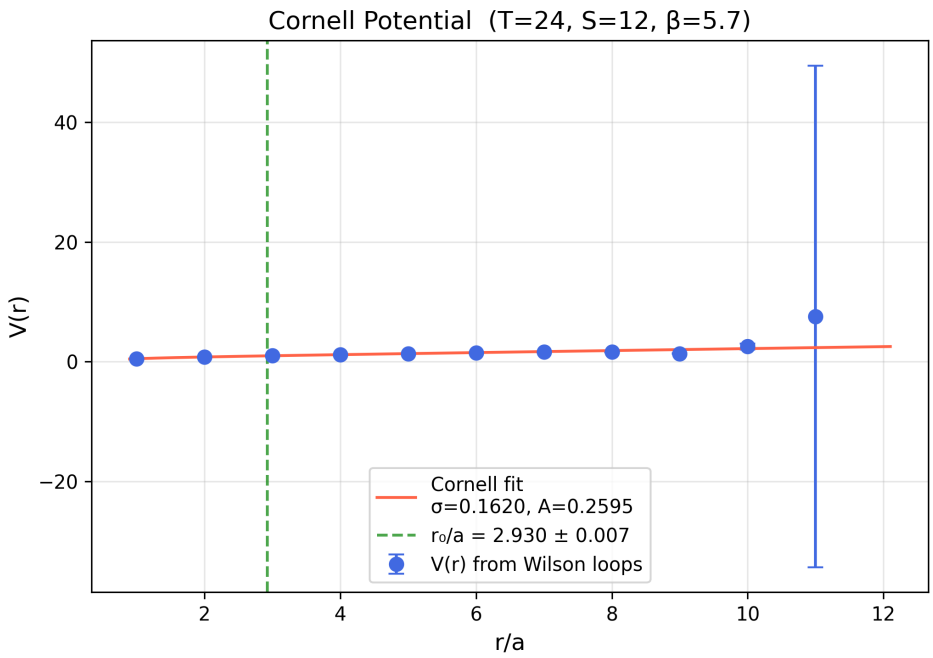


Figure 16: Cornell potential fit for 24×12^3 at $\beta = 5.7$

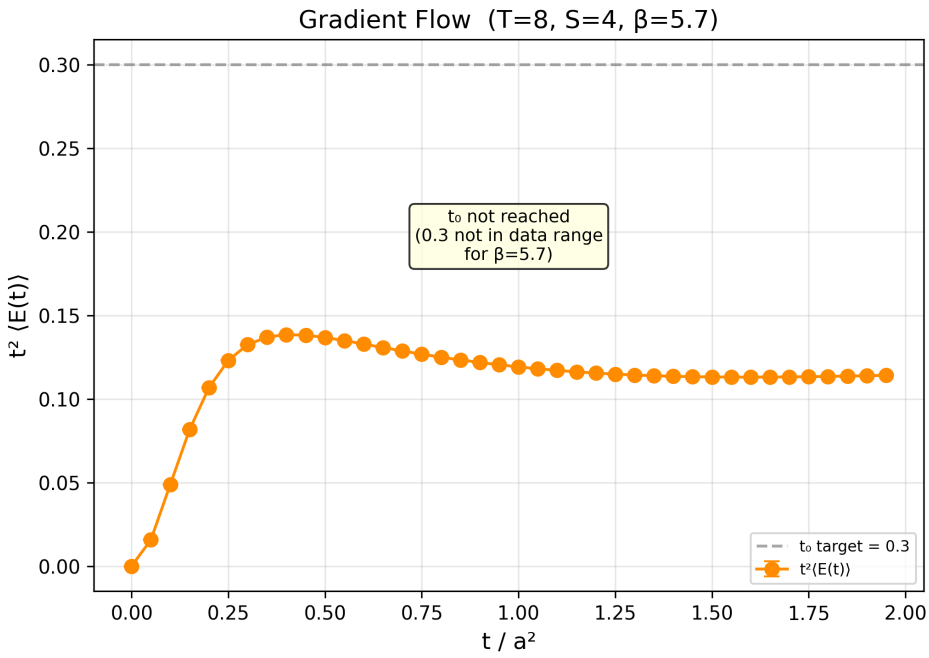


Figure 17: Gradient flow scale for 8×4^3 at $\beta = 5.7$

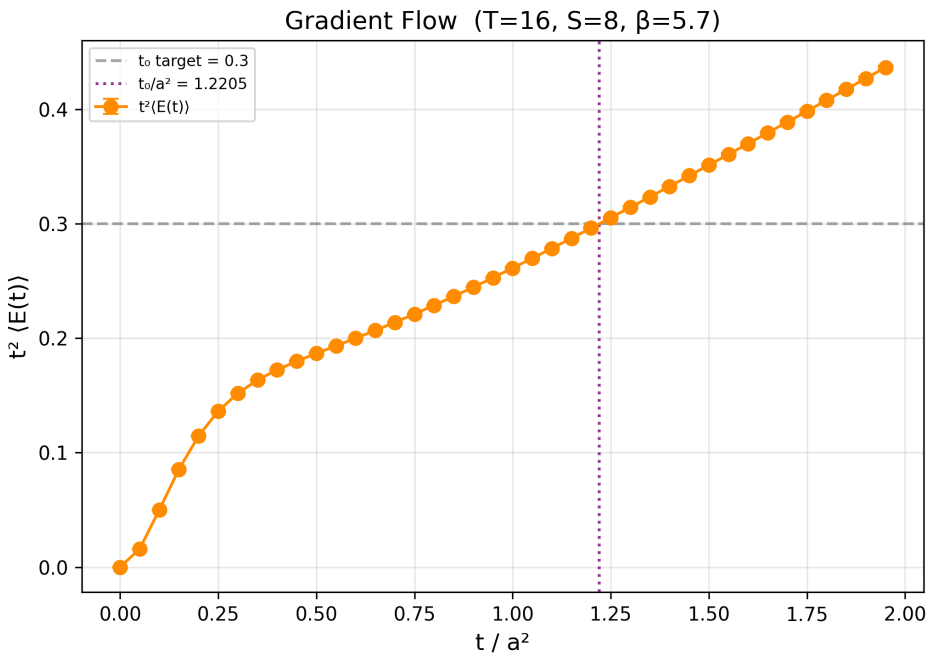


Figure 18: Gradient flow scale for 16×8^3 at $\beta = 5.7$

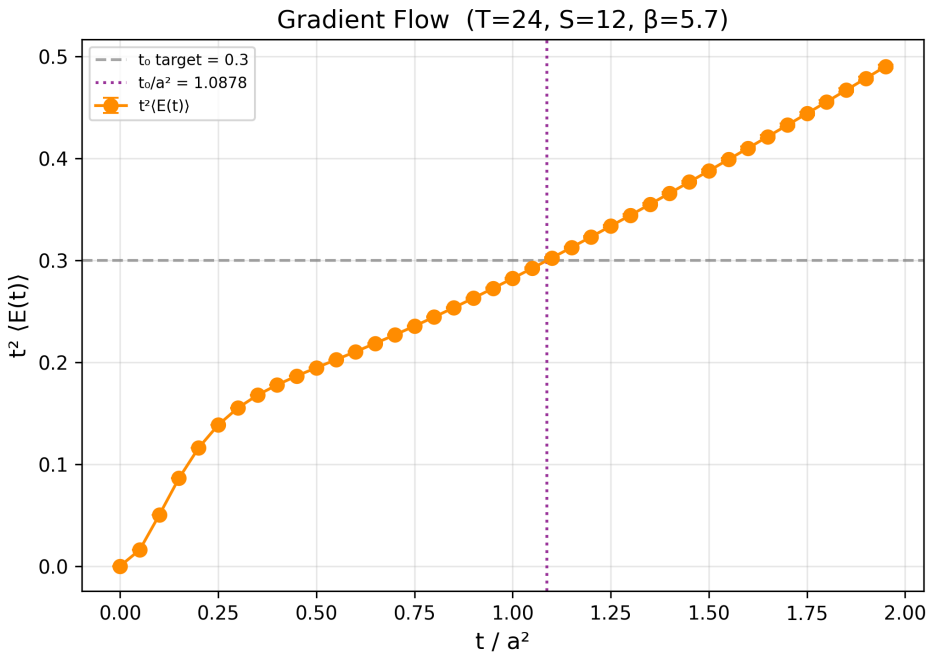


Figure 19: Gradient flow scale for 24×12^3 at $\beta = 5.7$

Beta Dependence at Fixed Volume (16×8^3)

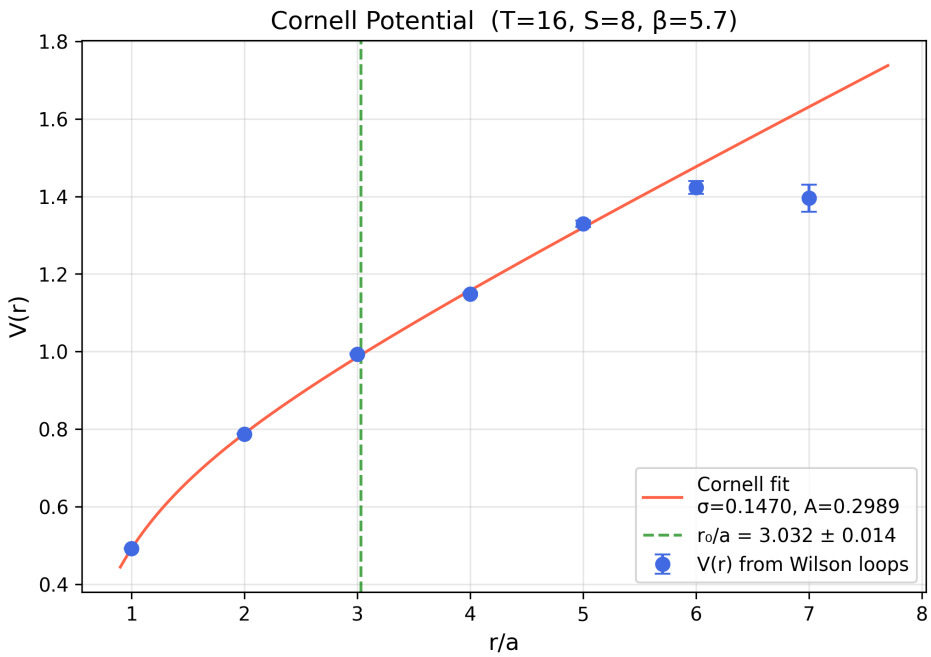


Figure 20: Cornell potential fit for $\beta = 5.7$

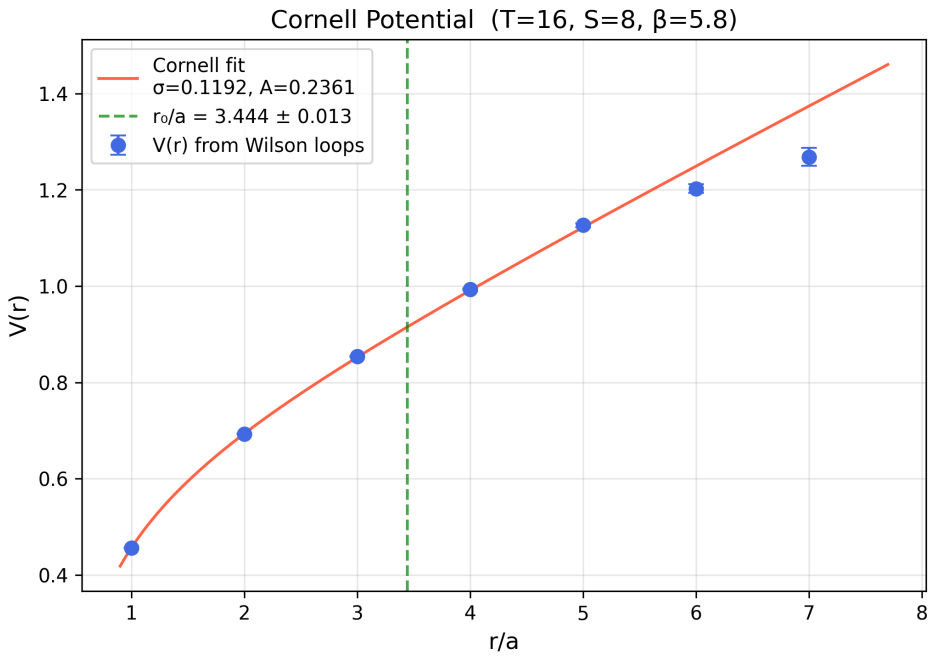
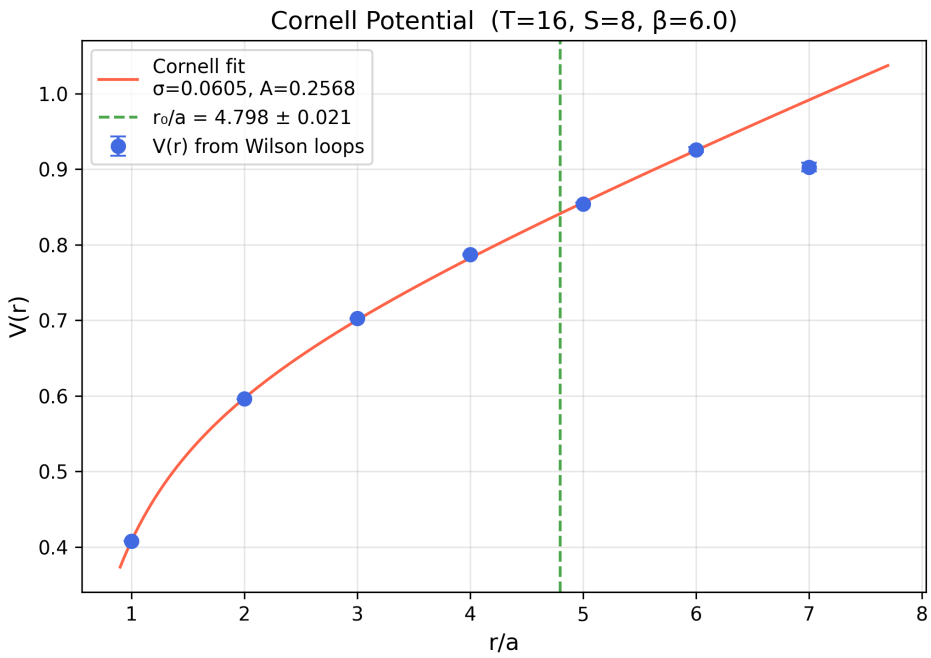


Figure 21: Cornell potential fit for $\beta = 5.8$



captionCornell potential fit for $\beta = 6.0$

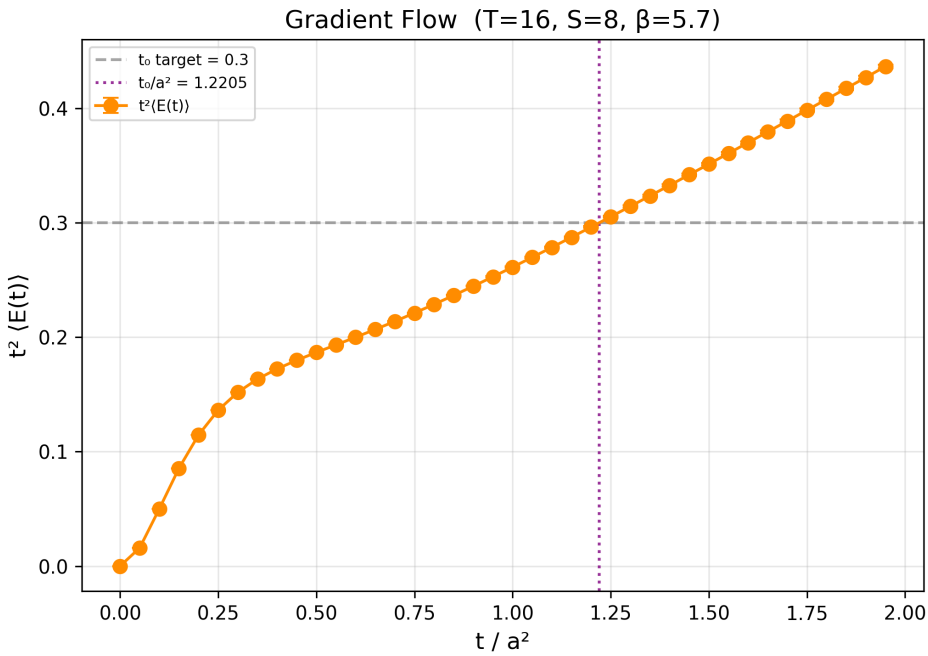


Figure 22: Gradient flow scale for $\beta = 5.7$

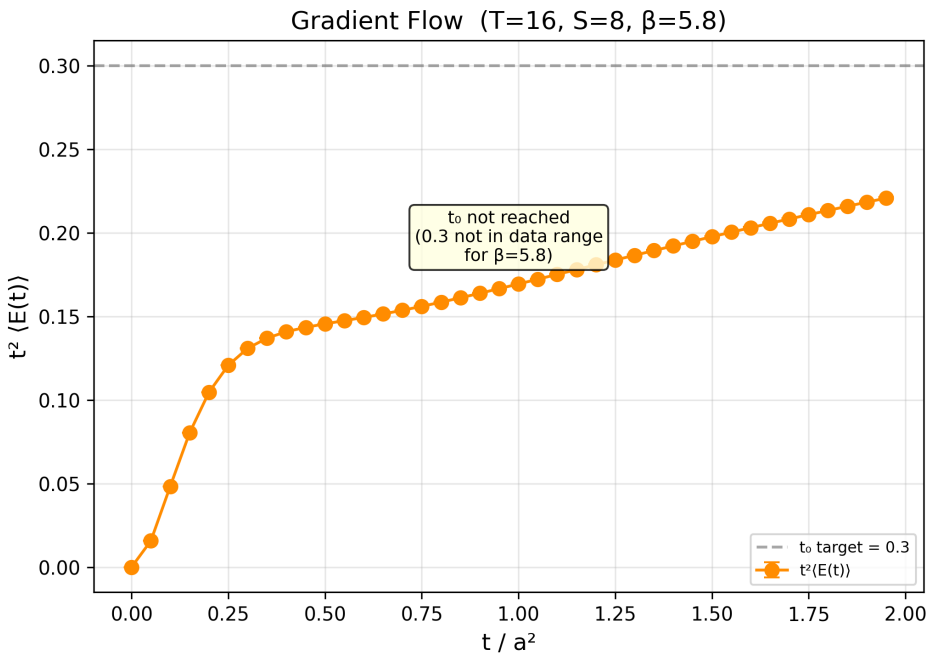


Figure 23: Gradient flow scale for $\beta = 5.7$

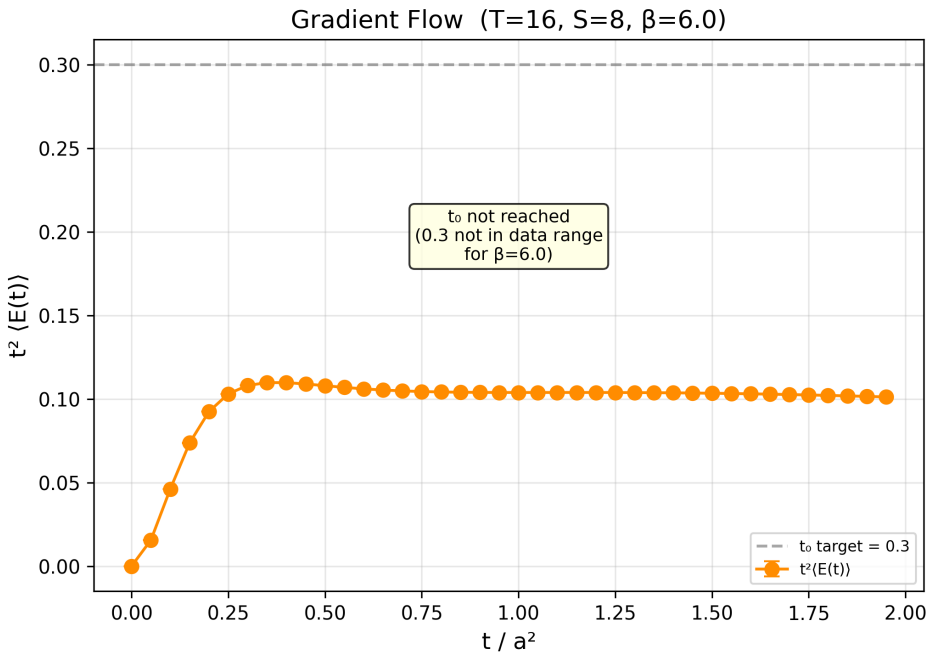


Figure 24: Gradient flow scale for $\beta = 5.7$

As expected, increasing β decreases the lattice spacing a. And because my lattice is too small when beta goes to 6.0, gradient flow radius may exceed size L and this method corrupts.

Topological Charge Evolution

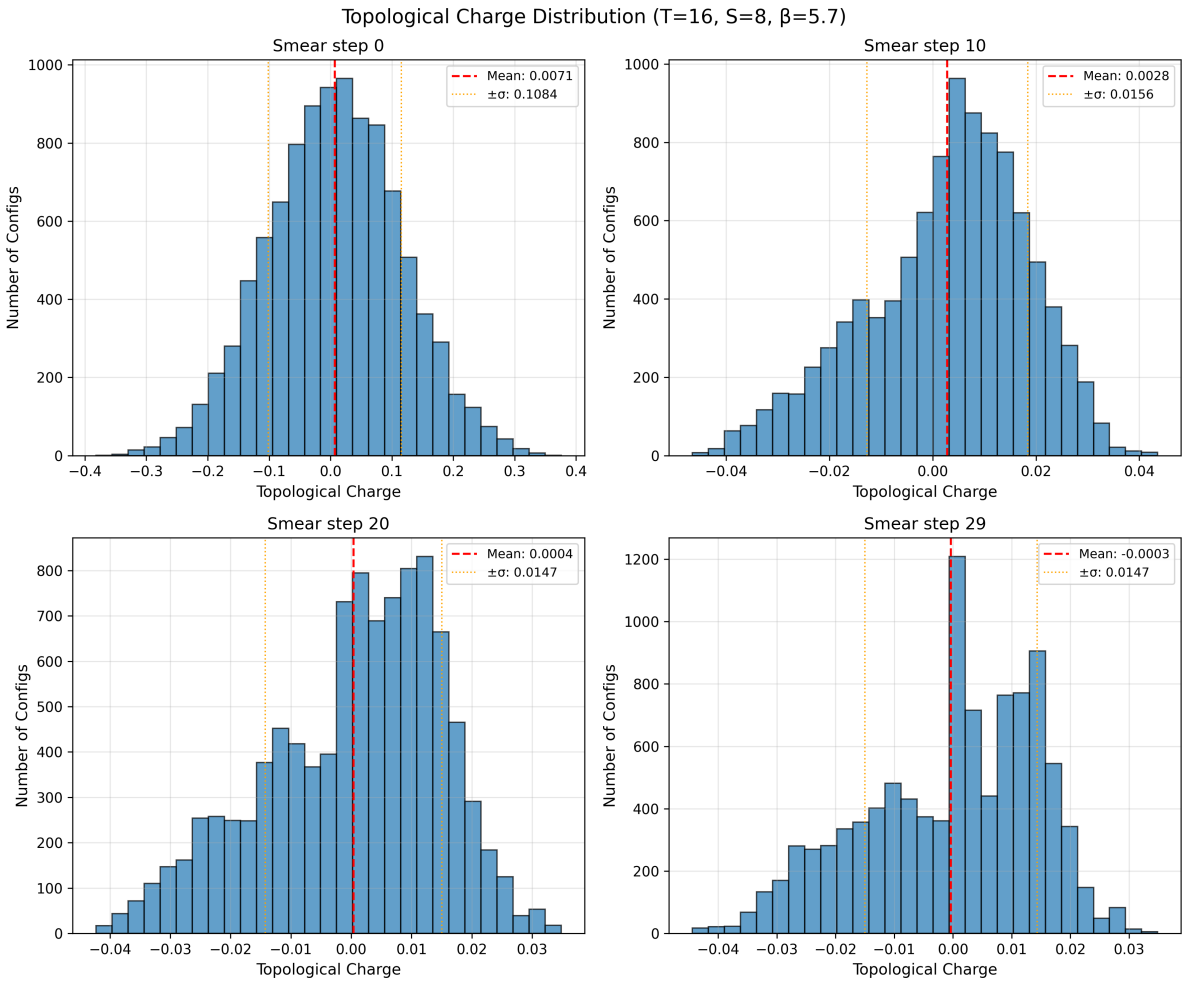


Figure 25: Topological charge Q distribution for 16×8^3 at $\beta = 5.7$

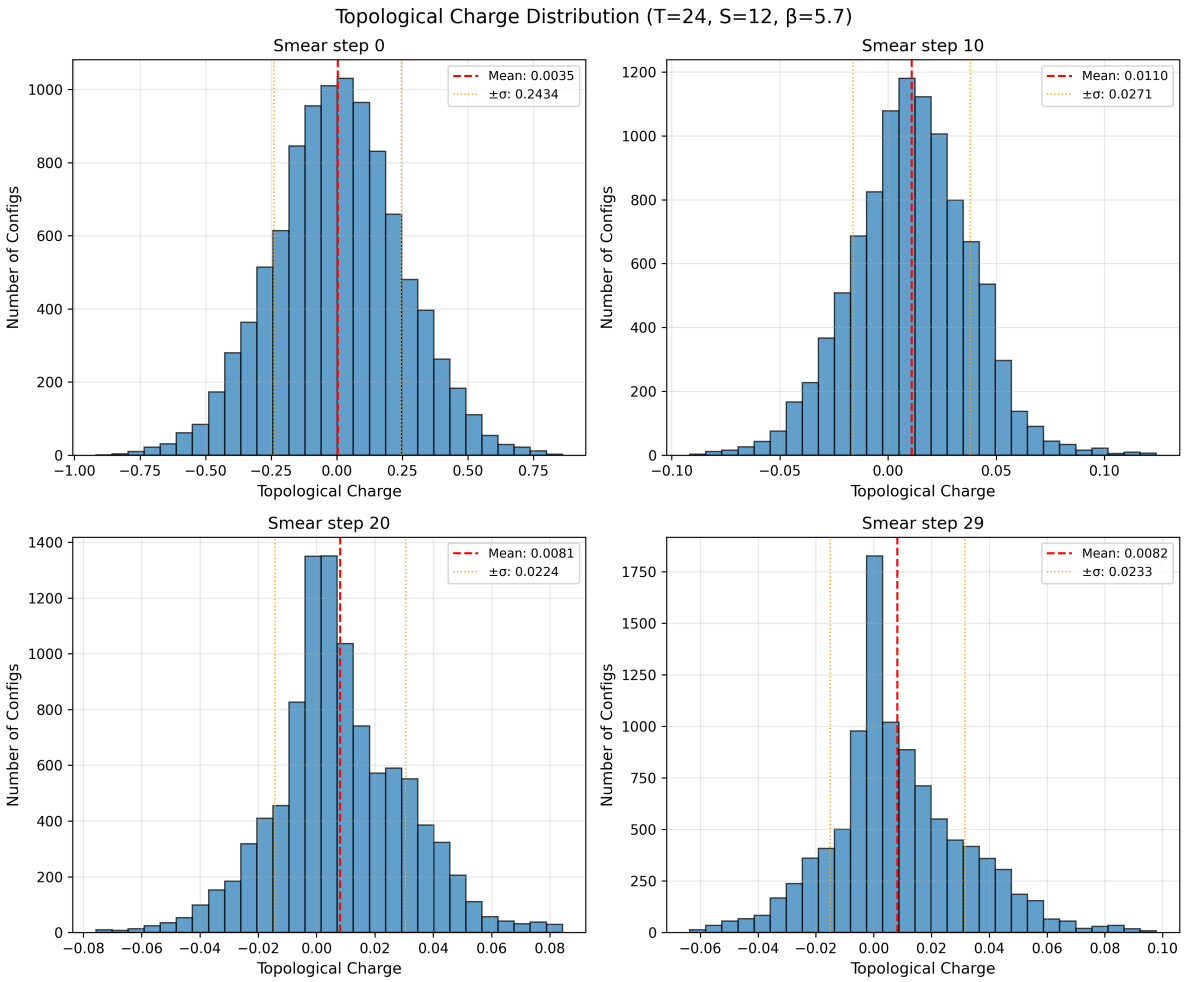


Figure 26: Topological charge Q distribution for 24×12^3 at $\beta = 5.7$

14. Definition of Topological Charge

Reconstruct topology

$$Tr(1 - U(p))_{max} < \epsilon \tag{29}$$

Following Lüscher's work (Commun. Math. Phys. 85, 39 (1982)), the lattice topological charge is defined using the clover-leaf construction of the field strength tensor. The lattice topological charge is:

$$Q = -\frac{1}{24\pi^2} \sum_{n\in\Lambda} \sum_{\mu,\nu,\rho,\sigma} \epsilon_{\mu\nu\rho\sigma} \Bigg\{ \int_{p(n,\mu,\nu)} d^2x \, \mathrm{Tr}\Big[\left(v_{n,\mu}\, \partial_\rho v_{n,\mu}^{-1}\right) \left(v_{n-\hat{\mu},\nu}\, \partial_\sigma v_{n-\hat{\mu},\nu}^{-1}\right) \Big] + \int_{f(n,\mu)} d^3x \, \mathrm{Tr}\Big[\left(v_{n,\mu}^{-1}\, \partial_\rho v_{n,\mu}\right) \left(v_{n,\mu}^{-1}\, \partial_\sigma v_{n,\mu}\right) \left(v_{n,\mu}^{-1}\, \partial_\sigma v_{n,\mu}\right) \Big] \Bigg\}. \tag{30}$$

In continuum QCD, the topological charge is defined through the field strength tensor $F_{\mu\nu}$ as:

$$Q = \frac{1}{32\pi^2} \int d^4x \, \varepsilon^{\mu\nu\rho\sigma} \mathrm{Tr}\left[F_{\mu\nu}(x) F_{\rho\sigma}(x)\right] \tag{31}$$

where $C_{\mu\nu}(x)$ is the clover field strength tensor defined as the average of four oriented plaquettes at site x:

$$C_{\mu\nu}(x_0) = \frac{1}{4} \sum_{x=x_0-\mu, x_0-\nu, x_0, x_\mu-\nu} P_{\mu\nu}(x) \tag{32}$$

Here $P_{\mu\nu}(x)$ is the plaquette:

$$P_{\mu\nu}(x) = U_\mu(x) U_\nu(x+\hat{\mu}) U_\mu^\dagger(x+\hat{\nu}) U_\nu^\dagger(x) \tag{33}$$

and the sum runs over all four orientations $(\pm\mu, \pm\nu)$ with appropriate signs to form the clover pattern.

The physical (traceless) field strength tensor is then:

$$F_{\mu\nu}(x) = Im[C_{\mu\nu}(x)] \tag{34}$$